

Pathology-Level ML Model Performance Summary (Test Set) Classical ML models trained on neural network pathology predictions							
Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra1	A Lines	Random Forest	0.710	[0.692-0.728]	43689	0.472	5
Attention Pool Extra1	A Lines	Logistic Regression	0.751	[0.737-0.766]	43689	0.472	5
Attention Pool Extra1	A Lines	XGBoost	0.755	[0.742-0.767]	43689	0.472	5
Attention Pool Extra1	Large Consolidation	Random Forest	0.824	[0.801-0.847]	43689	0.110	5
Attention Pool Extra1	Large Consolidation	Logistic Regression	0.884	[0.871-0.897]	43689	0.110	5
Attention Pool Extra1	Large Consolidation	XGBoost	0.871	[0.850-0.892]	43689	0.110	5
Attention Pool Extra1	Pleural Effusion	Random Forest	0.655	[0.624-0.685]	43689	0.011	5
Attention Pool Extra1	Pleural Effusion	Logistic Regression	0.785	[0.738-0.832]	43689	0.011	5
Attention Pool Extra1	Pleural Effusion	XGBoost	0.815	[0.773-0.856]	43689	0.011	5
Attention Pool Extra1	Other Pathology	Random Forest	0.605	[0.592-0.617]	43689	0.405	5
Attention Pool Extra1	Other Pathology	Logistic Regression	0.655	[0.639-0.671]	43689	0.405	5
Attention Pool Extra1	Other Pathology	XGBoost	0.645	[0.640-0.650]	43689	0.405	5
Attention Pool Extra	A Lines	Random Forest	0.710	[0.683-0.737]	43689	0.472	5
Attention Pool Extra	A Lines	Logistic Regression	0.749	[0.732-0.765]	43689	0.472	5
Attention Pool Extra	A Lines	XGBoost	0.755	[0.738-0.771]	43689	0.472	5
Attention Pool Extra	Large Consolidation	Random Forest	0.819	[0.788-0.850]	43689	0.110	5
Attention Pool Extra	Large Consolidation	Logistic Regression	0.883	[0.868-0.897]	43689	0.110	5
Attention Pool Extra	Large Consolidation	XGBoost	0.867	[0.849-0.885]	43689	0.110	5
Attention Pool Extra	Pleural Effusion	Random Forest	0.664	[0.562-0.767]	43689	0.011	5
Attention Pool Extra	Pleural Effusion	Logistic Regression	0.782	[0.731-0.834]	43689	0.011	5
Attention Pool Extra	Pleural Effusion	XGBoost	0.818	[0.778-0.859]	43689	0.011	5
Attention Pool Extra	Other Pathology	Random Forest	0.608	[0.600-0.617]	43689	0.405	5
Attention Pool Extra	Other Pathology	Logistic Regression	0.654	[0.645-0.662]	43689	0.405	5
Attention Pool Extra	Other Pathology	XGBoost	0.648	[0.638-0.658]	43689	0.405	5
Attention Pool	A Lines	Random Forest	0.687	[0.604-0.769]	43689	0.472	5
Attention Pool	A Lines	Logistic Regression	0.742	[0.673-0.810]	43689	0.472	5
Attention Pool	A Lines	XGBoost	0.727	[0.656-0.799]	43689	0.472	5
Attention Pool	Large Consolidation	Random Forest	0.784	[0.668-0.901]	43689	0.110	5
Attention Pool	Large Consolidation	Logistic Regression	0.875	[0.817-0.934]	43689	0.110	5
Attention Pool	Large Consolidation	XGBoost	0.828	[0.725-0.931]	43689	0.110	5
Attention Pool	Pleural Effusion	Random Forest	0.675	[0.651-0.698]	43689	0.011	5
Attention Pool	Pleural Effusion	Logistic Regression	0.739	[0.715-0.763]	43689	0.011	5
Attention Pool	Pleural Effusion	XGBoost	0.821	[0.790-0.853]	43689	0.011	5
Attention Pool	Other Pathology	Random Forest	0.590	[0.534-0.647]	43689	0.405	5
Attention Pool	Other Pathology	Logistic Regression	0.630	[0.617-0.643]	43689	0.405	5
Attention Pool	Other Pathology	XGBoost	0.627	[0.570-0.683]	43689	0.405	5